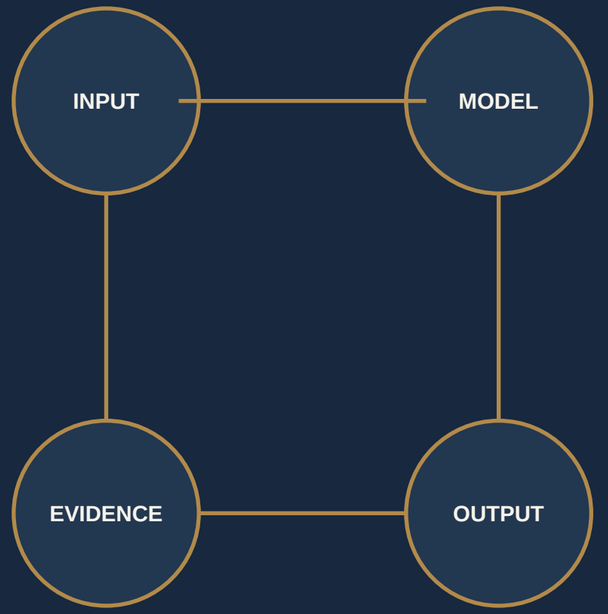
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Ethics

AI.402 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define ethical issues specifically associated with autonomous robots.
- Apply key methods including safety.
- Build or analyze a practical system involving responsibility.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Ethical issues specifically associated with autonomous robots

02

METHODS

Safety

03

SYSTEMS

Responsibility

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Ethical issues specifically as

Ethical issues specifically associated with autonomous robots

02 · Safety

Safety

03 · Responsibility

Responsibility

04 · Military applications

Military applications

05 · Surveillance

Surveillance

06 · Healthcare

Healthcare

07 · Employment and human-robo

Employment and human-robot relationships

08 · Mandatory practical: Perform d

Mandatory practical: Perform detailed ethical risk assessments on realistic robotic scenarios

Core ideas, made explicit

AI.402 • Distinguish definitions, assumptions and evidence

- Ethical issues specifically associated with autonomous robots
- Safety
- Responsibility
- Military applications
- Surveillance

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.402 • Distinguish definitions, assumptions and evidence

- Healthcare
- Employment and human-robot relationships
- Mandatory practical: Perform detailed ethical risk assessments on realistic robotic scenarios

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving ethical issues specifically associated with autonomous robots.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using ethical issues specifically associated with autonomous robots.
- Implement or simulate a baseline using safety.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.